

Bryan Bee

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EXPERIENCE

Technology Analyst — AWS Software Engineer

Sep. 2025 – Present

Accenture

Los Angeles, CA

- Developed the real-time event-routing layer for an AI agent-assist platform for Cigna that was approved to move forward with full production build-out following a stakeholder demo, using 2 AWS AppSync Events API channels to deliver AI-generated suggestions and Amazon Connect Contact Lens transcripts to customer service agents.
- Engineered and unit-tested Python event-publishing modules using AWS SigV4 authentication and AppSync Events APIs, enabling customer service agents to receive AI-generated guidance and live call-transcript data in real time.

Software Engineer Intern

Feb. 2025 – Jul. 2025

Leidos

Los Angeles, CA

- Designed and built a serverless ETL pipeline for 40+ datasets using AWS S3, Glue, and Athena to transform and structure large-scale datasets into analytics-ready fact and dimension tables, supporting a team of 6 data analysts.
- Created a RAG-based candidate matching platform using AWS Bedrock (Llama 3), vector embeddings, and semantic search to retrieve relevant resumes and generate grounded candidate-job evaluations, automating 300+ candidate-job matches and reducing manual resume screening.

Software Engineer Intern

Feb. 2024 – May 2024

Partillion Bioscience

Pasadena, CA

- Developed a computer vision pipeline using Python and OpenCV for nano-vial particle detection, improving quality control success rates by 30%.
- Automated merging of composite color channels for 400+ nano-vial images per week using Python, enhancing images used in cancer studies and eliminating hours of manual compositing work.

Software Engineer Intern

Jun. 2023 – Aug. 2023

Northrop Grumman

Baltimore, MD

- Automated metric generation in Python using the Jira REST API, pandas, Matplotlib, and NumPy to pull and calculate engineering metrics, automating manual metrics collection and reducing collection effort by 90% across projects while powering 50% of the automation platform's reporting metrics.

EDUCATION

California State University, Los Angeles

Los Angeles, CA

Bachelor of Science in Computer Science, GPA: 3.9

PROJECTS

AdServe — Real-Time Ad Serving & Auction Platform

Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, Kafka, Docker, Locust

- Engineered a real-time ad auction platform modeling 2M users, 5K advertisers, and ~23K campaigns, supporting audience targeting, weighted ranking, budget enforcement, and impression/click/conversion tracking.
- Optimized the auction hot path by eliminating SQLAlchemy N+1 queries, reducing request latency $\sim 9\times$ ($1.75s \rightarrow 0.15\text{--}0.28s$); root-caused a Redis cache stampede under concurrent load and redesigned caching with stale-while-revalidate, cutting median latency $3.1s \rightarrow 94ms$ ($\sim 33\times$) and increasing throughput $2.8\times$ over the uncached baseline.
- Scaled the FastAPI service with multi-process workers, connection-pool tuning, and concurrency profiling to ~ 770 RPS with 300ms p95 latency and 0% failures at 150 concurrent users, while sustaining 1.1K+ RPS with 0% failures in stress tests up to 1,600 concurrent users.
- Eliminated a concurrent lost-update race that silently dropped 90% of campaign charges by implementing atomic PostgreSQL updates; reduced the write path from 7 to 3 SQL statements ($\sim 57\%$ fewer round trips).
- Built an idempotent Kafka event pipeline with at-least-once delivery for click/conversion processing and user-preference learning; integrated learned interest weights into auction ranking, producing a measured $\sim 26\%$ relative increase in wins for ads matching users' most-engaged interests.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML, CSS, Kotlin

Cloud/Databases: AWS (S3, Lambda, EC2, Glue, Athena, Bedrock, AppSync, Amazon Connect, CDK), PostgreSQL, MySQL, DynamoDB, Firebase Firestore, Redis

Developer Tools: Git, Docker, Locust

Libraries/Frameworks: FastAPI, Flask, SQLAlchemy, React.js, pandas, Kafka, NumPy, Matplotlib, Bootstrap